A Skilled Secret Sharing Scheme for *r*-Uniform Hypergraph-Based Prohibited Structures[1]

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Abstract

Secret sharing is that a dealer distributes a piece of information (called a share) about a secret to each participant such that authorized subsets of participants can reconstruct the secret but unauthorized subsets of participants cannot determine the secret. If any unauthorized subset of participants can not obtain any more information regarding the secret, then the scheme is called perfect. A hypergraph is a general graph in which each edge may be formed by more than two vertices. A prohibited structure is a family of the subset of participants such that any subset belongs to this set can not reconstruct the secret. In this paper, we propose a new perfect secret sharing schemes for prohibited structure based on r-uniform hypergraph where a vertex denotes a participant and an edge is a set of participants who cannot recover the secret. The information rate of our schemes is $max\{r/(C(n-1, r-1) - d + r + 1), r/(C(n-1, r-1) - d + 1)\}$*, where n is the number of participants and d is the minimum degree of the hypergraph. We will show this scheme is perfect and, makes the scheme of Sun to be a special case. Besides, we also show that this scheme is better than TUM scheme when given a same prohibited structure.*

1. Introduction

Secret sharing schemes deal with the problem of how to securely share a *secret* K among a set of participants P. In a secret sharing scheme, a secret is broken into pieces by a *dealer*, called *shares*, where each participant keeps some shares. When a *qualified* subset of participants put their shares together, they can recover the secret K, but any unqualified subset of participants can not. The collection of qualified subsets is called *access structure* (denoted by Γ); the collection of unqualified subsets is called *prohibited structure* (denoted by Δ) . In 1979, Blakley [1] and Shamir [15] independently introduced the concept of secret sharing. In Shamir's (t, n)-threshold scheme [15], every t participants can recover the secret K, but any sets of less than t participants cannot get any information about the secret. Hence $\Gamma = \{A\text{: } |A| \geq t \text{ and } A \subseteq P\}$ and $\Delta = \{A\text{: } |A| < t \text{ and } A \subseteq P\}$.

If any set in the prohibited structure not only can not recover the secret, but also can not know as much information regarding the secret as they have no any shares, then the scheme is called *perfect* [6][8][10][15]. Let $P = \{p_1, p_2, \cdots, p_n\}$ be the set of all participants and 2^P be the set of all subsets of P. Let $\mathcal{K}$ be the *secret space* and $\mathcal{S}$ be the *share space*. In general, the efficiency of a secret sharing scheme is measured by the *information rate* ρ [3] defined as

$$\rho = \min\{\rho_i : 1 \leq i \leq n\}, \rho_i = \frac{\log|\mathcal{K}|}{\log|\mathcal{S}_i|},$$

where $\mathcal{S}_i$ denotes the set of possible shares that p_i might receive. If a secret sharing scheme is practical, we do not went to have to distribute too much secret information as shares. Consequently, we want to make the information rate as high as possible. A secret sharing scheme is *ideal* if $\rho = 1$. For any qualified subset $A \in \Gamma$, any superset of A is also an qualified subset intuitively. Hence, the access structure should satisfy the *monotone increasing property*:

$$A \in \Gamma \text{ and } A \subseteq B \subseteq P \Rightarrow B \in \Gamma.$$

Similarly, for any unqualified subset $A \in \Delta$, any subset of A is also an unqualified subset intuitively. Hence, the prohibited structure satisfies the *monotone decreasing property*:

$$A \in \Delta \text{ and } B \subseteq A \subseteq P \Rightarrow B \in \Delta.$$

Let Γ_0 be a family of the minimal sets in Γ, called the *minimal access structure*. Γ_0 is denoted by $\Gamma_0 = \{A \in \Gamma\text{: } A' \not\subset A \text{ for all } A' \in \Gamma - \{A\}\}$. The family of maximal set in Δ, called *maximal prohibited structure*, is denoted by Δ_1. That is, $\Delta_1 = \{B \in \Delta_1\text{: } B \not\subset B' \text{ for all } B' \in \Delta - \{B\}\}$.

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In 1997, Sun and Shieh [20][21] classified secret sharing scheme into the following three types according access structure or prohibited structure:

Type I: *access structure* Γ

For every secret sharing scheme with an access structure Γ in type I, only subset of participants in access structure can recover the secret K, others can not. So that prohibited structure $\Delta = 2^P \setminus \Gamma$ is implied.

Type II: *prohibited structure* Δ

For every secret sharing scheme with a prohibited structure Δ in type II, only subsets of participants in prohibited structure Δ can not recover the secret K, others can. So that access structure $\Gamma = 2^P \setminus \Delta$ is implied.

Type III: *mixed structure* (Γ, Δ)

For every secret sharing scheme with a mixed structure (Γ, Δ) in type III, subsets of participants in access structure Γ can recover the secret, subsets of participants in prohibited structure Δ can not recover the secret K, and $\Gamma \cap \Delta = \phi$ is implied. The other subsets which are not in Γ and Δ may either can recover the secret K or not. The set $2^P \setminus(\Gamma \cup \Delta)$ can be empty or not.

Let G be a graph and $v \in V$. The number of edges incident at v in G is called the *degree* of the vertex v in G and is denoted by $d_G(v)$. In a graph, each vertex can be view as a participant; each edge is a relation of two participants by that the secret K can be recovered or not [3][20]. Brickell and Stinson [3] studied perfect secret sharing schemes for graph-based access structure Γ. In *graph-based access structure* Γ, the access structure contains the pairs of participants corresponding to edges. In 1994, Shieh and Sun [17] studied perfect secret sharing schemes for graph-based mixed structure (Γ, Δ). In *graph-based mixed structure* (Γ, Δ), Γ contains the pairs of participants corresponding to edges of the graph and Δ contains the pairs of participants corresponding to non-edges of the graph. In 1996, they further studied secret sharing schemes for graph-based prohibited structure [20]. In *graph-based prohibited structures* Δ, the prohibited structure Δ contains every participant and the pairs of participants which corresponded to edges. The access structure Γ is the pairs of participants corresponding to non-edges and the union of $\{A: A \subseteq P \text{ and } |A| \geq 3\}$.

In 1997, Sun and Shieh proposed a construction for perfect secret sharing schemes with graph-based prohibited structures [20][21]. By the construction of their proposed, if there are n participants, the information rate of the constructed perfect secret sharing scheme is $2/n$. In 1999, Sun further improved the information rate of the constructed perfect secret sharing scheme to be max $\{2/(\deg(\overline{G}) + 3), 2/n\}$ by given another scheme, where $\deg(\overline{G})$ is the maximum *degree* of the complement of G [19]. The structures are considered as "graph-based", and every edge in a graph is adjacent by two vertices. Hence the graph-based prohibited structures always consider the special prohibited structure that for any set $A \in \Delta$, the size of A is equal to 2 or 1.

A *hypergraph* H is an ordered pair (V, E) where V is a non-empty set of vertices and $E = \{E_1, E_2,\ldots, E_m\} \subseteq 2^V$ where $|E_i| \geq 2$ for $1 \leq i \leq m$ is a set of *hyperedges*. The degree of a vertex u is $|\{E_i: u \in E_i \text{ for } 1 \leq i \leq m\}|$ and denote this number by $\deg(u)$. The maximum vertex degree among all vertices of hypergraph H is denoted by $\deg(H)$, and the minimum one is denoted by $d(H)$. If $|E_i| = r \geq 2$ for all $1 \leq i \leq |E|$, the hypergraph H is called an *r-uniform hypergraph*. Given an r-uniform hypergraph $H = (P, S)$ such that $P = \{p_1, p_2, \ldots, p_n\}$ is the set of vertices of the hypergraph H corresponding to the participants of a secret sharing scheme. Let S denote the set of r participants corresponding to hyperedges in H; R denote the set of r participants who are not in S. That is, $S = E(H)$ and $R = \{A: A \notin E(H) \text{ and } |A| = r\}$. Similarity, we define the prohibited structure for the reason of satisfying monotone. That is, when giving a r-uniform hypergraph H, the prohibited structure is denoted by $\Delta = \{A: A \subseteq P \text{ and } |A| \leq (r-1)\} \cup \{A: A \in S\}$, and then the access structure is decided by $\Gamma = 2^P \setminus \Delta = \{A: A \subseteq P \text{ and } |A| \geq (r+1)\} \cup \{A: A \in R\}$. It is easily to check that the prohibited structure and the access structure are monotone. Hence, we will discuss the prohibited structure based on r-uniform hypergraph in which for any set $A \in \Delta$, the size of A is not always l or 2.

Throughout this paper, p and q denote large prime such that q divides $p - 1$. Let Z_q be a finite field with q elements. Let the generator g be an element of order q of multiplicative group Z_q^*.

In Section 2, we first give some knowledge and previous results that will help us to propose our new scheme. In Section 3, we propose a secret sharing scheme based on r-uniform hypergraph prohibited structure, called r-HP scheme. In Section 4, we compare our scheme with other schemes. After that, we give a conclusion in Section 5.

2. Preliminaries

In this section, we list a scheme and a function that will be used to construct our scheme in the following of this paper.

2.1 Shamir's (*t*, *n*)-threshold scheme

The access structure Γ of (t, n)-threshold scheme is described as $\Gamma = \{A: |A| \geq t$ and $A \in 2^P\}$; the prohibited structure Δ is described as $\Delta = \{A: |A| < t$ and $A \in 2^P\}$; the number of participants is n. Shamir's (t, n)-threshold scheme is perfect and described as follows [15]:

There is a finite field Z_q in where q is a large prime and assumes key space $\mathcal{K}$ and share space $\mathcal{S}$ are over Z_q.

Step 1: A dealer chooses n distinct nonzero elements over the finite field Z_q which is denoted by $x_1, x_2, \ldots, x_n$ and values x_i are public for all $1 \leq i \leq n$.

Step 2: Suppose the dealer wants to share a secret key K, and then the dealer chooses $t - 1$ elements $a_1, a_2, \ldots, a_{t-1}$ over Z_q independently with the uniform distribution.

Step 3: Let $f(x) = a_{t-1}x^{t-1} + \ldots + a_2x^2 + a_1x + K \pmod q$ be a polynomial of degree $t - 1$ over the Z_q. The dealer distributes the share $S_i = f(x_i)$ to participant p_i for all $1 \leq i \leq n$.

Obviously, given any t shares $S_{i_1}, S_{i_2}, \ldots, S_{i_t}$, $\{i_1, i_2, \ldots, i_t\} \subset \{1, 2, \ldots, n\}$, $f(x)$ can be reconstructed from the Lagrange interpolating polynomial as follows [6]:

$$f(x) = \sum_{j=1}^{t} (S_{i_j} \cdot \prod_{m=1, m\neq j}^{t} \frac{(x - i_m)}{(i_j - i_m)}) \pmod q.$$

Thus, the secret K can be obtained by $f(0)$.

2.2 One-way hash-function

A *one-way hash-function* is a one-way function $g_{\text{hash}}(x)$ which satisfy the following properties:

(1) g_{hash} maps bit-strings of arbitrary length to bit-strings of a fixed length.

(2) When given g_{hash} and an input x, $g_{\text{hash}}(x)$ is easy to compute.

(3) To find any pre-image x' such that $g_{\text{hash}}(x') = y$ when given any y for which a corresponding input is not known.

(4) It is computationally infeasible to find any second input which has the same output as any specified input.

(5) It is computationally infeasible to find any two distinct inputs x, x' which hash to the same output.

The properties have been proven in [9].

3. New scheme for *r*-uniform hypergraph-based prohibited structure

In this section, we propose the new scheme when $r = 3$ as beginning and analyze the security of this scheme in Sub-section 3.2. After that, there is an example in Sub-section 3.3. Finally, we will extend the scheme for r-uniform hypergraph in Sub-section 3.4 and analyze the information rate at least.

3.1 3-HP scheme

Given a 3-uniform hypergraph H for the prohibited structure, we constructed a perfect secret sharing scheme as follows. Assume that $P = \{p_1, p_2, \ldots, p_n\}$ is the set of participants corresponding to the vertices of the hypergraph H. $S = \{A: A \in E(H)\}$ and $R = \{A: A \notin E(H)$ and $|A| = 3\}$. The prohibited structure is denoted by $\Delta = \{A: A \subseteq P$ and $|A| \leq 2\} \cup \{A: A \in S\}$, and the access structure is decided by $\Gamma = 2^P \setminus \Delta = \{A: A \subseteq P$ and $|A| \geq 4\} \cup \{A: A \in R\}$. We define T is the set $\{p_i: \exists\, p_j, p_k$, and p_l in P such that $\{p_i, p_j, p_k\}, \{p_i, p_j, p_l\}, \{p_i, p_k, p_l\}, \{p_j, p_k, p_l\} \in E(H)$ and $1 \leq i \neq j \neq k \neq l \leq n\}$ and use the symbol $\oplus$ to express operation of the exclusive or. The scheme is described as follows:

We use the one-way hash-function g_{hash} and assume that all computations are over Z_q where $q \geq \max\{K_0, K_1, K_2\}$ is a large prime.

Step 1: Let the secret $K = (K_0, K_1, K_2)$ be taken randomly from $[Z_q]^3$. A dealer D constructs $f(x) = K_2x^2 + K_1x + K_0$. Let $y_{(i,j)}$ be computed from $f(x)$ as follows: $y_{(i,j)} = f(i \cdot n + j)$, for $1 \leq i < j \leq n - 1$, and $y_j = y_{(0,j)} = f(j)$ for $j = 1, 2$, and 3.

Step 2: D construct three Shamor's $(4, n)$-threshold schemes, named TS_1, TS_2, and TS_3, to protect y_1, y_2 and y_3. For each $(4, n)$-TS_i, let y_i be its sub-secret and $s_{i,1}, s_{i,2}, \ldots, s_{i,n}$ be its n sub-shares.

Step 3: D selects n random numbers, $r_1, r_2, \ldots, r_n$, over Z_q. Then the share of participant p_i is given by $S_i = \langle a_{i,(1,2)}, a_{i,(1,3)}, \ldots, a_{i,(j,l)}, \ldots, a_{i,(n-1,n)}, a_{i,1}, a_{i,2}, a_{i,3}, a_{i,4}\rangle$, where $1 \leq j < l \leq n$:

$$a_{i,(j,k)} = \begin{cases} y_{(i,j)} + g_{\text{hash}}(r_j \oplus r_k), & \text{if } i \notin \{j,k\} \text{ and } \\ & \{p_i, p_j, p_k\} \notin E(H); \\ \text{empty}, & \text{otherwise.} \end{cases}$$

$$a_{i,j} = \begin{cases} s_{j,i}, & \text{if } p_i \in T \text{ for } j = 1,2,3; \\ \text{empty}, & \text{otherwise.} \end{cases}$$

$a_{i,4} = r_i$.

Next, we will show that the constructed secret sharing scheme satisfies the following four conditions:

(1) if $A \subseteq P$ and $|A| \le 2$, then A obtains no information regarding the secret;

(2) if $A \in S$, then A obtains no information regarding the secret;

(3) if $A \in R$, then A can recover the secret;

(4) if $A \subseteq P$ and $|A| \ge 4$, then A can recover the secret.

We prove these four conditions for 3-HP scheme in Sub-section 3.2.

3.2 Security analysis for 3-HP scheme

In this sub-section, we analyze the security of the 3-HP scheme by Theorem 3.1 to Theorem 3.4.

Theorem 3.1.
For all $A \subseteq P$ and $|A| \le 2$, A obtains no information regarding the secret of the 3-HP scheme.

Proof. We discuss in the following two cases:

Case 1:
if $|A| = 1$ and assume $A = \{p_i\}$, where $1 \le i \le n$. S_i is the shares of p_i and $S_i = \langle a_{i,(1,2)}, a_{i,(1,3)}, \ldots, a_{i,(n-1,n)}, a_{i,1}, a_{i,2}, a_{i,3}, a_{i,4}\rangle$. It is clear that no information about $y_{(j,k)}$ for all $0 \le j < k \le n$, can be derived from S_i. Therefore, participant p_i can not obtain any information of secret key K.

Case 2:
if $|A| = 2$ and assume $A = \{p_i, p_j\}$, where $1 \le i < j \le n$. The share of $S_i = \langle a_{i,(1,2)}, a_{i,(1,3)}, \ldots, a_{i,(n-1,n)}, a_{i,1}, a_{i,2}, a_{i,3}, a_{i,4}\rangle$ is the shares of p_i; the share of $S_j = \langle a_{j,(1,2)}, a_{j,(1,3)}, \ldots, a_{j,(n-1,n)}, a_{j,1}, a_{j,2}, a_{j,3}, a_{j,4}\rangle$ is the shares of p_j. It is clear that no information about $y_{(k,l)}$ for all $0 \le k < l \le n$, can be derived from S_i and S_j. Therefore, participants p_i and p_j can not obtain any information of secret key K. □

Theorem 3.2.
For all $A \in S$, A obtains no information regarding the secret key of the 3-HP scheme.

Proof. Assume that $A = \{p_i, p_j, p_k\}$, where $1 \le i < j < k \le n$. $S_t = \langle a_{t,(1,2)}, a_{t,(1,3)}, \ldots, a_{t,(n-1,n)}, a_{t,1}, a_{t,2}, a_{t,3}, a_{t,4}\rangle$ is the shares of p_t for all $t \in \{i, j, k\}$. Because $A \in S$, $\{p_i, p_j, p_k\} \in E(H)$. The only shares that A possible to obtain is $y_{i,j}$, $y_{i,k}$ and $y_{j,k}$. But the participant p_i owns $a_{i,4} = r_i$, $a_{i,(i,j)}$, $a_{i,(i,k)}$ and $a_{i,(j,k)}$ are all empty; the participant p_j owns $a_{j,4} = r_j$, $a_{j,(i,j)}$, $a_{j,(j,k)}$ and $a_{j,(i,k)}$ are all empty; the participant p_k owns $a_{k,4} = r_k$, $a_{k,(i,k)}$, $a_{k,(j,k)}$ and $a_{k,(i,j)}$ are all empty, too. Therefore, they can not obtain any information on $y_{(i,j)}$, $y_{(i,k)}$ and $y_{(j,k)}$ actually. So, A can not obtains any information regarding the secret. □

Theorem 3.3.
For all $A \in R$, A can recover the secret key of the 3-HP scheme.

Proof. Assume that $A = \{p_i, p_j, p_k\}$, where $1 \le i < j < k \le n$. The share of $S_t = \langle a_{t,(1,2)}, a_{t,(1,3)}, \ldots, a_{t,(n-1,n)}, a_{t,1}, a_{t,2}, a_{t,3}, a_{t,4}\rangle$ is the shares of p_t for all $t \in \{i, j, k\}$. Because $A \in R$, $\{p_i, p_j, p_k\} \notin E(H)$. Hence participant p_i owns $a_{i,4} = r_i$ and $a_{i,(j,k)} = y_{(j,k)} + g(r_j \oplus r_k)$; participant p_j owns $a_{j,4} = r_j$ and $a_{j,(i,k)} = y_{(i,k)} + g(r_i \oplus r_k)$; participant p_k owns $a_{k,4} = r_k$ and $a_{k,(i,j)} = y_{(i,j)} + g(r_i \oplus r_j)$. Therefore, they can recover $y_{(i,j)}$, $y_{(i,k)}$ and $y_{(j,k)}$. After that, they can determine $f(x)$ and hence the secret K. □

Theorem 3.4.
For all $A \subseteq P$ and $|A| \ge 4$, A can recover the secret key of the 3-HP scheme.

Proof. Assume $A = \{p_i, p_j, p_k, p_l\}$, where $1 \le i < j < k < l \le n$. If there exists three participants of A belong to R, the secret key can be recovered from Theorem 2.3. All we need to consider is the case that all $\{p_i, p_j, p_k\}$, $\{p_i, p_j, p_l\}$, $\{p_i, p_k, p_l\}$ and $\{p_j, p_k, p_l\}$ are edges of H. If so, then p_i, p_j, p_k and $p_l \in T$. Therefore, participant p_t owns $a_{t,1} = s_{1,t}$, $a_{t,2} = s_{2,t}$, and $a_{t,3} = s_{3,t}$ for all $t \in \{i, j, k, l\}$. Hence p_i, p_j, p_k and p_l can recover y_1, y_2 and y_3, and so do the secret key K. □

3.3 An example for 3-HP scheme

In this section, we give an example by using our method. There is a 3-uniform hypergraph $H = (V, E)$ in Figure 1 which denotes the prohibited structure with six participants. Let $P = \{p_1, p_2, p_3, p_4, p_5, p_6\}$. $E(H) = \{\{p_1, p_2, p_3\}, \{p_1, p_2, p_4\}, \{p_1, p_3, p_4\}, \{p_2, p_3, p_4\}, \{p_3, p_4, p_5\}\} = S$, and $R = \{\{p_1, p_2, p_5\}, \{p_1, p_2, p_6\}, \{p_1, p_3, p_5\}, \{p_1, p_3, p_6\}, \{p_1, p_4, p_5\}, \{p_1, p_4, p_6\}, \{p_1, p_5, p_6\}, \{p_2, p_3, p_5\}, \{p_2, p_3, p_6\}, \{p_2, p_4, p_5\}, \{p_2, p_4, p_6\}, \{p_2, p_5, p_6\}, \{p_3, p_4, p_6\}, \{p_3, p_5, p_6\}, \{p_4, p_5, p_6\}\}$. The prohibited structure $\Delta = \{\phi, \{p_1\}, \{p_2\}, \{p_3\}, \{p_4\}, \{p_5\}, \{p_6\}, \{p_1, p_2\}, \{p_1, p_3\}, \{p_1, p_4\}, \{p_1, p_5\}, \{p_1, p_6\}, \{p_2, p_3\}, \{p_2, p_4\}, \{p_2, p_5\}, \{p_2, p_6\}, \{p_3, p_4\}, \{p_3, p_5\}, \{p_3, p_6\}, \{p_4, p_5\}, \{p_4, p_6\}, \{p_5, p_6\}\} \cup S$. The access structure $\Gamma = R \cup \{A\colon A \subseteq P \text{ and } |A| \ge 4\}$.

Let the secret key $K = (K_0, K_1, K_2)$ is taken randomly from $Z_q \times Z_q \times Z_q$, where q is a large prime. Let $f(x) = K_2x^2 + K_1x + K_0 \pmod q$. $y_{(i,j)}$ is computed from $f(x)$ as: $y_{(i,j)} = f(i \cdot n + j) \pmod q$, for $0 \le i < j \le 6$ and $y_j = y_{(0,j)}$. Next, we construct three conventional (4, 6)-threshold schemes, named TS_1, TS_2, and TS_3, to protect y_1, y_2 and y_3. Let $SK_1 = y_1$, $SK_2 = y_2$, and $SK_3 = y_3$; $s_{1,1}, s_{1,2}, \ldots, s_{1,6}, s_{2,1}, s_{2,2}, \ldots, s_{2,6}$ and $s_{3,1}, s_{3,2}, \ldots,$

$s_{3,6}$ be their six sub-shares, respectively. Thus, given any four sub-shares, $s_{i,j}$, $s_{i,k}$, $s_{i,l}$, $s_{i,m}$($1 \le j < k < l < m \le n$), the sub-secret key SK_i can be recovered, but less than four sub-shares provide no information about SK_i for $1 \le i \le 3$. A function g is a one way function and number r_1, r_2, …, r_6 are selected randomly over Z_q. The shares of participants are given by Table 1, where "–" denotes empty entry: In the following, we demonstrate the constructed secret sharing schemes satisfies:

(1) if $A \subseteq P$ and $|A| \le 2$, then A obtains no information regarding the secret,

(2) if $A \in S$, then A obtains no information regarding the secret,

(3) if $A \in R$, then A can recover the secret,

(4) if $A \subseteq P$ and $|A| \ge 4$, then A can recover the secret.

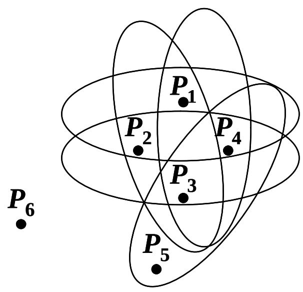

Figure 1. An example with six participants

For (1), given $A = \{p_5, p_6\} \in \Delta$, A can not obtain any information on $y_{(5,6)}$ (actually, all shares $y_{(i,j)}$). Therefore, A obtains no information about the secret K.

For (2), given $A = \{p_1, p_2, p_3\} \in \Delta$, A can not obtain any information on $y_{(1,2)}$, $y_{(1,3)}$ and $y_{(2,3)}$ (and all the other $y_{(i,j)}$). Therefore, A obtains no information about the secret K.

For (3), given $A = \{p_1, p_2, p_5\} \in \Gamma$, A can recover the secret K because participants p_1, p_2 and p_4 can recover $y_{(2,5)}$, $y_{(1,2)}$, and $y_{(1,5)}$ and hence the polynomial $f(x)$.

For (4), given $A = \{p_1, p_2, p_3, p_4\} \in \Gamma$, A can recover the secret K because participants p_1, p_2, p_3 and p_4 can recover y_1, y_2 and y_3 and hence the polynomial $f(x)$.

3.4 General *r*-HP scheme

Given a r-uniform hypergraph H for the prohibited structure, we constructed a perfect secret sharing scheme as follows. Assume that $P = \{p_1, p_2, \dots, p_n\}$ is the set of participants corresponding to the vertices of the hypergraph H. S to denote the set of r-subset of participants corresponding to edges in H; R to denote the set of r-subset of participants corresponding to not an edge in H. That is, $S = \{A: A \in E(H)\}$ and $R = \{A: A \notin E(H) \text{ and } |A| = r\}$. It is reasonable to restrict that the prohibited structure and the access structure are monotone. When giving a r-uniform hypergraph H, the prohibited structure is denoted by $\Delta = \{A: A \subseteq P \text{ and } |A| \le r-1\} \cup \{A: A \in S\}$, and then the access structure is decided by $\Gamma = 2^P \setminus \Delta = \{A \mid A \subseteq P \text{ and } |A| \ge r+1\} \cup \{A: A \in R\}$. We define T is the set $\{p_i: \exists\, p_{j_1}, p_{j_2}, \dots, p_{j_r}$ such that $\{p_{j_1}, p_{j_2}, \dots, p_{j_r}\} \in E(H)$, and $\{p_i, p_{j_1}, p_{j_2}, \dots p_{j_{k-1}}, p_{j_{k+1}}, \dots, p_{j_r}\} \in E(H)$ for all $1 \le k \le r\}$.We use the symbol $\oplus$ to express operation of the exclusive or. The scheme is described as follows:

We use the one-way hash-function g_{hash} and assume that all computations are over Z_q.

Step 1: Let the secret $K = (K_0, K_1, \dots, K_{r-1})$ be taken randomly from $[Z_q]^r$, and let $f(x) = K_{r-1}x^{r-1} + K_{r-2}x^{r-2} + \dots + K_1x + K_0 \pmod q$. $y_{(i_1,i_2,\dots,i_{r-1})}$ is computed from $f(x)$ as: $y_{(i_1,i_2,\dots,i_{r-1})} = f(\sum_{k=1}^{r-1} i_k n^{r-k-1})$, for $0 \le i_1 \le i_2 \le \dots \le i_{r-1} \le n$ and $y_j = y_{(0,0,\dots,0,j)}$ for $1 \le j \le r$.

Step 2: The dealer D choose r^2 elements $b_{i,1}$, $b_{i,2}$, …, $b_{i,r}$ over Z_q and construct r Shamir's $(r+1, n)$-threshold schemes named TS_i for $1 \le i \le r$. Let $TS_i(x) = b_{i,r}x^r + b_{i,r-1}x^{r-1} + \dots + b_{i,1}x + y_i \pmod q$ for $1 \le i \le r$. For each $(r+1, n)$-TS_i, let y_i be its sub-secret and $s_{i,1}$, $s_{i,2}$, …, $s_{i,n}$ be its n sub-shares where $s_{i,k} = TS_i(k)$ for $1 \le k \le n$.

Step 3: D selects n random numbers, r_1, r_2, …, r_n, over Z_q. The share of participant p_i is given by $S_i = \langle a_{i,(1,2,\dots,r-1)}, \dots, a_{i,(j1,\ j2,\dots,\ jr-1)}, \dots, a_{i,(n-r+2,\dots,n)}, a_{i,1}, \dots, a_{i,r+1}\rangle$, for $1 \le j_1 < j_2 < \dots < j_{r-1} \le n$, where

$$a_{i,(j_1,j_2,\dots,j_{r-1})} = \begin{cases} y_{(j_1,j_2,\dots,j_{r-1})} + g_{\text{hash}}(r_{j_1} \oplus r_{j_2} \oplus \dots \oplus r_{j_{r-1}}), & \text{if } i \notin \{j_1,\dots,j_{r-1}\} \text{ and } \{p_i, p_{j_1}, p_{j_2}, \dots, p_{j_{r-1}}\} \notin E(H); \\ \text{empty}, & \text{otherwise.} \end{cases}$$

$$a_{i,j} = \begin{cases} s_{j,i}, & \text{if } p_i \in T \text{ for } 1 \le j \le r; \\ \text{empty}, & \text{otherwise.} \end{cases}$$

$a_{i,r+1} = r_i$.

Obviously, given $y_{(i_1,i_2,\dots,i_{r-1})}$ and $y_{(j_1,j_2,\dots,j_{r-1})}$, for all $1 \le k \le r-1$, $i_k \ne j_k$, $f(x)$ can be determined uniquely. Therefore, one who gets r or more $y_{(i_1,i_2,\dots,i_{r-1})}$s can recover the secret key K. However, one without knowledge of any $y_{(i_1,i_2,\dots,i_{r-1})}$ obtain no information about the secret key K. Note that one who gets any one $y_{(i_1,i_2,\dots,i_{r-1})}$ can obtain partial information about the secret key. Given any $r+1$ sub-shares, s_{i,j_1}, s_{i,j_2},

Table 1: The Shares of Six Participants

	S_1	S_2	S_3	S_4	S_5	S_6
$a_{i,(1,2)}$	–	–	–	–	–	$y_{(1,2)} + g_{\text{hash}}(r_1 \oplus r_2)$
$a_{i,(1,3)}$	–	–	–	–	$y_{(1,3)} + g_{\text{hash}}(r_1 \oplus r_3)$	$y_{(1,3)} + g_{\text{hash}}(r_1 \oplus r_3)$
$a_{i,(1,4)}$	–	–	–	–	$y_{(1,4)} + g_{\text{hash}}(r_1 \oplus r_4)$	$y_{(1,4)} + g_{\text{hash}}(r_1 \oplus r_4)$
$a_{i,(1,5)}$	–	$y_{(1,5)} + g_{\text{hash}}(r_1 \oplus r_5)$	$y_{(1,5)} + g_{\text{hash}}(r_1 \oplus r_5)$	$y_{(1,5)} + g_{\text{hash}}(r_1 \oplus r_5)$	–	$y_{(1,5)} + g_{\text{hash}}(r_1 + r_5)$
$a_{i,(1,6)}$	–	$y_{(1,6)} + g_{\text{hash}}(r_1 \oplus r_6)$	$y_{(1,6)} + g_{\text{hash}}(r_1 \oplus r_6)$	$y_{(1,6)} + g_{\text{hash}}(r_1 \oplus r_6)$	$y_{(1,6)} + g_{\text{hash}}(r_1 \oplus r_6)$	–
$a_{i,(2,3)}$	–	–	–	–	$y_{(2,3)} + g_{\text{hash}}(r_2 \oplus r_3)$	$y_{(2,3)} + g_{\text{hash}}(r_2 \oplus r_3)$
$a_{i,(2,4)}$	–	–	–	–	$y_{(2,4)} + g_{\text{hash}}(r_2 \oplus r_4)$	$y_{(2,4)} + g_{\text{hash}}(r_2 \oplus r_4)$
$a_{i,(2,5)}$	$y_{(2,5)} + g_{\text{hash}}(r_2 \oplus r_5)$	–	$y_{(2,5)} + g_{\text{hash}}(r_2 \oplus r_5)$	$y_{(2,5)} + g_{\text{hash}}(r_2 \oplus r_5)$	–	$y_{(2,5)} + g_{\text{hash}}(r_2 \oplus r_5)$
$a_{i,(2,6)}$	$y_{(2,6)} + g_{\text{hash}}(r_2 \oplus r_6)$	–	$y_{(2,6)} + g_{\text{hash}}(r_2 \oplus r_6)$	$y_{(2,6)} + g_{\text{hash}}(r_2 \oplus r_6)$	$y_{(2,6)} + g_{\text{hash}}(r_2 \oplus r_6)$	–
$a_{i,(3,4)}$	–	–	–	–	–	$y_{(3,4)} + g_{\text{hash}}(r_3 \oplus r_4)$
$a_{i,(3,5)}$	$y_{(3,5)} + g_{\text{hash}}(r_3 \oplus r_5)$	$y_{(3,5)} + g_{\text{hash}}(r_3 \oplus r_5)$	–	–	–	$y_{(3,5)} + g_{\text{hash}}(r_3 \oplus r_5)$
$a_{i,(3,6)}$	$y_{(3,6)} + g_{\text{hash}}(r_3 \oplus r_6)$	$y_{(3,6)} + g_{\text{hash}}(r_3 \oplus r_6)$	–	$y_{(3,6)} + g_{\text{hash}}(r_3 \oplus r_6)$	$y_{(3,6)} + g_{\text{hash}}(r_3 \oplus r_6)$	–
$a_{i,(4,5)}$	$y_{(4,5)} + g_{\text{hash}}(r_4 \oplus r_5)$	$y_{(4,5)} + g_{\text{hash}}(r_4 \oplus r_5)$	–	–	–	$y_{(4,5)} + g_{\text{hash}}(r_4 \oplus r_5)$
$a_{i,(4,6)}$	$y_{(4,6)} + g_{\text{hash}}(r_4 \oplus r_6)$	$y_{(4,6)} + g_{\text{hash}}(r_4 \oplus r_6)$	$y_{(4,6)} + g_{\text{hash}}(r_4 \oplus r_6)$	–	$y_{(4,6)} + g_{\text{hash}}(r_4 \oplus r_6)$	–
$a_{i,(5,6)}$	$y_{(5,6)} + g_{\text{hash}}(r_5 \oplus r_6)$	$y_{(5,6)} + g_{\text{hash}}(r_5 \oplus r_6)$	$y_{(5,6)} + g_{\text{hash}}(r_5 \oplus r_6)$	$y_{(5,6)} + g_{\text{hash}}(r_5 \oplus r_6)$	–	–
$a_{i,1}$	$s_{1,1}$	$s_{1,2}$	$s_{1,3}$	$s_{1,4}$	–	–
$a_{i,2}$	$s_{2,1}$	$s_{2,2}$	$s_{2,3}$	$s_{2,4}$	–	–
$a_{i,3}$	$s_{3,1}$	$s_{3,2}$	$s_{3,3}$	$s_{3,4}$	–	–
$a_{i,4}$	r_1	r_2	r_3	r_4	r_5	r_6

..., $s_{i,j_{r+1}}$ ($1 \le j_1 < j_2 < ... < j_{r+1} \le n$), the sub-master key y_i can be recovered, but less than $r + 1$ sub-shares provide no information about y_i. Next, we will show that the constructed secret sharing scheme satisfies the following four conditions:

(1) if $A \subseteq P$ and $|A| \le r - 1$, then A obtains no information regarding the secret;

(2) if $A \in S$, then A obtains no information regarding the secret;

(3) if $A \in R$, then A can recover the secret;

(4) if $A \subseteq P$ and $|A| \ge r + 1$, then A can recover the secret.

So we have the following four theorems in Sub-section 3.5.

3.5 Security analysis and information rate for *r*-HP scheme

In this sub-section, we prove that r-HP scheme is perfect by Theorem 3.5 to Theorem 3.8 and analyze the information rate of this scheme by Theorem 3.9 to Corollary 3.11.

Theorem 3.5.
Suppose $A \subseteq P$ and $|A| \le r - 1$, A obtains no information regarding the secret of the r-HP scheme.

Theorem 3.6.
Suppose $A \in S$, A obtains no information regarding the secret of the r-HP scheme.

Theorem 3.7.
Suppose $A \in R$, A can recover the secret of the r-HP scheme.

Theorem 3.8.
Suppose $A \subseteq P$ and $|A| \ge r + 1$, A can recover the secret of the r-HP scheme.

The proofs of Theorem 3.5 to Theorem 3.8 are the same as Theorem 3.1 to Theorem 3.4, respectively. So we skip it. According to Theorem 3.5 to Theorem 3.8 and we only used the perfect (t, n)-threshold scheme in the proposed scheme, so our scheme is also perfect.

Theorem 3.9.
Suppose n is the number of vertices of H and d is the minimum degree of H. The information rate ρ of the r-HP scheme satisfies:

1) $r/(C(n - 1, r - 1) - d + r + 1) \le \rho \le r/ (C(n - 1, r - 1) - d + 1)$ if $d \ge r$,

2) $\rho \ge r/(C(n - 1, r - 1) + 1)$.

Proof. The share S_i of participant p_i is an $(C(n, r - 1) + r + 1)$-dimensional vector. For the first $C(n, r - 1)$ dimensions, if $\{p_{l_1}, p_{l_2}, \ldots, p_{l_r}\} \in E(H)$ where $1 \le l_1$

$< l_2 < \ldots < l_r \le n$, $a_{l_i,(x_1,\ldots,x_{r-1})}$ are empty for any $l_i \in \{x_1, x_2, ..., x_{r-1}\} \subseteq \{1, 2, ..., n\}$ or $\{x_1, x_2, ..., x_{r-1}\} = \{l_1, l_2, ..., l_r\} - \{l_i\}$. Otherwise, $a_{l_i,(x_1,\ldots,x_{r-1})}$s are over Z_q. For the last $r + 1$ dimensions, $a_{l_i,r+1}$ is r_i always; $a_{l_i,j}$ are either over $GF(q^3)$ or empty according to p_i is belongs to T or not for $1 \le j \le r$. Therefore, the size of share Sl_i is at most $\log(q^{C(n-1,r-1)-\deg(p_{l_i})+r+1})$ (when $a_{l_i,j}$, are non-empty for $1 \le j \le r$) and at least $\log(q^{C(n-1,r-1)-\deg(p_{l_i})+1})$ (when $a_{li,j}$, are empty for $1 \le j \le r$), where $\deg(p_{l_i})$ is the degree of vertex p_{l_i} in the hypergraph H. Hence, the maximal size of the shares is at most $\log(q^{C(n-1,\ r-1)-d+4})$ and at least $\log(q^{C(n-1,\ r-1)-d+1})$, where d is the minimum degree of H. And because the size of the secret key is $\log(q^r)$, the information rate of the perfect secret sharing scheme will satisfies $(r \times \log q)/((C(n-1, r-1) - d + r + 1) \times \log q) \le \rho \le (r \times \log q)/((C(n-1, r-1) - d + 1) \times \log q)$, as 1) demand.

To prove 2), note that whenever $a_{i,j}$ are non-empty for $1 \le j \le r$, such that $p_i, p_{l_1}, \ldots, p_{l_r}$ satisfy $\{p_{x_1}, p_{x_2}, .., p_{x_r}\} \in H$ where $\{x_1, x_2, ..., x_r\} \subseteq \{i, l_1, l_2, ..., l_r\}$. Thus $a_{i,(x_1,x_2,\ldots,x_r)}$ are empty. That means $\deg(p_i) \ge 3$. Hence $\rho \ge 3/(C(n-1, 2) + 1)$ is always holds. □

Theorem 3.10.
The r-uniform Hypergraph H has the information rate $\rho = r/(C(n-1, r-1) - d + 1)$ if $d < r$, where d is the minimum degree of H and n is the number of vertices of H.

Proof. Here we follow the proof of Theorem 3.9. We assume the share of participant p_i has the maximum size among all of shares. If $\deg(p_i) < r$, where $\deg(p_i)$ is the degree of vertex p_i of H, then we can show $p_i \notin T$ as follows. Assume $p_i \in T$, then there exists $p_{l_1}, p_{l_2}, \ldots, p_{l_r}$ where $1 \le l_1 < l_2 < \ldots < l_r \le n$ and $i \notin \{l_1, l_2, \ldots, l_r\}$ such that $\{p_i, p_{x_1}, p_{x_2}, \ldots, p_{x_{r-1}}\} \in E(H)$ where $\{x_1, \ldots, x_{r-1}\} \subseteq \{l_1, l_2, \ldots, l_r\}$. Thus the degree of vertex p_i of H is at least r. This contradicts the fact that $\deg(p_i) < r$. Hence, $p_i \notin T$. For the first $C(n, r-1)$ dimensions, there are $C(n-1, r-1) - d$ of $a_{i,(x1,\ldots,x_{r-1})}$'s which are non-empty for $1 \le x_1 < x_2 < \ldots < x_{r-1} \le n$, $i \notin \{x_1, x_2, \ldots, x_{r-1}\}$, when $d < 3$. For the last $r + 1$ dimensions, $a_{i,r+1}$ is non-empty; $a_{i,j}$ are empty for $1 \le j \le r$ because $p_i \notin T$. Therefore, the maximal size of shares is $\log(q^{C(n-1,r-1)-d+1})$ for $d = 0$, 1 or 2. The size of the secret key is $\log(q^r)$. Thus, the information rate of the constructed perfect secret sharing scheme is $\rho = r/(C(n-1, r-1) - d + 1)$ when $d < r$. □

From Theorem 3.9 and Theorem 3.10, we can obtain the following corollary.

Corollary 3.11.
The lower bound of the information rate of the r-HP scheme is $\max\{r/(C(n-1, r-1) - d + r + 1), r/(C(n-1, r-1) + 1)\}$, where d is the minimum degree of H and n is the number of vertices of H.

4. Compare *r*-HP scheme with other scheme

In 2005, Tochikubo, Uyematsu, and Matsumoto [14] propose an efficient scheme based on authorized subsets. We call this scheme TUM scheme. The TUM scheme is based on general access structure. In this section, we compare TUM scheme with our scheme based on r-uniform hypergraph-based prohibited structure. We first describe TUM scheme and give an example for 3-uniform hypergraph-based prohibited structure. And analysis the information rate of TUM scheme for r-uniform hypergraph-based prohibited structure.

4.1 TUM scheme

For $P = \{p_1, p_2, \ldots, p_n\}$, $K \in \mathcal{K}$ and Γ, the proposed scheme is described as follows.

Step 1: Let $\Gamma'_0 = \{A \in \Gamma_0\colon |A| \le l\}$, where $l = \max_{B\in\Delta} |B|$ and represent it as $\Gamma'_0 = \{A_1, A_2, \ldots, A_d\}$ with $d = |\Gamma'_0|$.

Step 2: Let $P' = \{P \in X\colon X \in \Gamma_0$ and $|X| > l\}$ and $n' = |P'|$. Compute n' shares $S = \{s_1, s_2, \ldots, s_{n'}\}$ for the secret K by using Shamir's $(l + 1, n')$-threshold scheme. Then, one distinct share in S is assigned to each participant in P'.

Step 3: For every $A_i \in \Gamma'_0$, compute $|A_i|$ shares $S_i = \{s_{n'+i,1}, s_{n'+i,2}, \ldots, s_{n'+i,|A_i|}\}$ by using Shamir's $(|A_i|, |A_i|)$-threshold scheme with K as a secret independently for $1 \le i \le d$. One distinct share in S_i is assigned to each $P \in A_i$ $(1 \le i \le d)$.

4.2 An example for 3-uniform hypergraph-based prohibited structure

In this sub-section, we give an example for . There is a 3-uniform hypergraph $H = (V, E)$ in Figure 1 which denotes the prohibited structure with six participants. The maximal prohibited structure $\Delta_1 = \{\{p_1, p_5\}, \{p_1, p_6\}, \{p_2, p_5\}, \{p_2, p_6\}, \{p_3, p_6\}, \{p_4, p_6\}, \{p_5, p_6\}, \{p_1, p_2, p_3\}, \{p_1, p_2, p_4\}, \{p_1, p_3, p_4\}, \{p_2, p_3, p_4\}, \{p_3, p_4, p_5\}\}$. The minimal access structure $\Gamma_0 = \{\{p_1, p_2, p_5\}, \{p_1, p_2, p_6\}, \{p_1, p_3, p_5\}, \{p_1, p_3, p_6\}, \{p_1, p_4, p_5\}, \{p_1, p_4, p_6\}, \{p_1, p_5, p_6\}, \{p_2, p_3, p_5\}, \{p_2, p_3, p_6\}, \{p_2, p_4, p_5\}, \{p_2, p_4, p_6\}, \{p_2, p_5, p_6\}, \{p_3, p_4, p_6\}, \{p_3, p_5, p_6\}, \{p_4, p_5, p_6\}, \{p_1, p_2, p_3, p_4\}\}$.

(1) Let Γ_0' = {A_1, A_2, A_3, A_4, A_5, A_6, A_7, A_8, A_9, A_{10}, A_{11}, A_{12}, A_{13}, A_{14}, A_{15}} = {{p_1, p_2, p_5}, {p_1, p_2, p_6},{ p_1, p_3, p_5}, {p_1, p_3, p_6}, {p_1, p_4, p_5}, {p_1, p_4, p_6}, {p_1, p_5, p_6}, {p_2, p_3, p_5}, {p_2, p_3, p_6}, {p_2, p_4, p_5}, {p_2, p_4, p_6}, {p_2, p_5, p_6}, {p_3, p_4, p_6}, {p_3, p_5, p_6}, {p_4, p_5, p_6}} and l = 3.

(2) Let P' = {p_1, p_2, p_3, p_4}. Compute 4 shares S = {s_1, s_2, s_3, s_4,} for the secret K by using Shamir's (4, 4)-threshold scheme. Then, one distinct share in S is assigned to each participant in P'.

(3) S_1 = {$s_{5,1}$, $s_{5,2}$, $s_{5,3}$}, S_2 = {$s_{6,1}$, $s_{6,2}$, $s_{6,3}$}, S_3 = {$s_{7,1}$, $s_{7,2}$, $s_{7,3}$}, S_4 = {$s_{8,1}$, $s_{8,2}$, $s_{8,3}$}, S_5 = {$s_{9,1}$, $s_{9,2}$, $s_{9,3}$}, S_6 = {$s_{10,1}$, $s_{10,2}$, $s_{10,3}$}, S_7 = {$s_{11,1}$, $s_{11,2}$, $s_{11,3}$}, S_8 = {$s_{12,1}$, $s_{12,2}$, $s_{12,3}$}, S_9 = {$s_{13,1}$, $s_{13,2}$, $s_{13,3}$}, S_{10} = {$s_{14,1}$, $s_{14,2}$, $s_{14,3}$}, S_{11} = {$s_{15,1}$, $s_{15,2}$, $s_{15,3}$}, S_{12} = {$s_{16,1}$, $s_{16,2}$, $s_{16,3}$}, S_{13} = {$s_{17,1}$, $s_{17,2}$, $s_{17,3}$}, S_{14} = {$s_{18,1}$, $s_{18,2}$, $s_{18,3}$}, S_{15} = {$s_{19,1}$, $s_{19,2}$, $s_{19,3}$}.

In this case, shares are distributed in Table 2. The information rate of this example is 1/10.

Table 2: The Shares of Six Participants by TUM Scheme

Participant p_i	The share of participant p_i
p_1	{s_1, $s_{5,1}$, $s_{6,1}$, $s_{7,1}$, $s_{8,1}$, $s_{9,1}$, $s_{10,1}$, $s_{11,1}$}
p_2	{s_2, $s_{5,2}$, $s_{6,2}$, $s_{12,1}$, $s_{13,1}$, $s_{14,1}$, $s_{15,1}$, $s_{16,1}$}
p_3	{s_3, $s_{7,2}$, $s_{8,2}$, $s_{12,2}$, $s_{13,2}$, $s_{17,1}$, $s_{18,1}$}
p_4	{s_4, $s_{9,2}$, $s_{10,2}$, $s_{14,2}$, $s_{15,2}$, $s_{17,2}$, $s_{19,1}$}
p_5	{$s_{5,3}$, $s_{7,3}$, $s_{9,3}$, $s_{11,2}$, $s_{12,3}$, $s_{14,3}$, $s_{16,2}$, $s_{18,2}$, $s_{19,2}$}
p_6	{$s_{6,3}$, $s_{8,3}$, $s_{10,3}$, $s_{11,3}$, $s_{13,3}$, $s_{15,3}$, $s_{16,3}$, $s_{17,3}$, $s_{18,3}$, $s_{19,3}$}

4.3 The information rate of TUM scheme for r-uniform hypergraph-based prohibited structure

In this section, we calculate the information rate of TUM scheme based on r-uniform hypergraph prohibited structure by Theorem 4.1 and Corollary 4.2.

Theorem 4.1.
Suppose n is the number of vertices of H and d is the minimum degree of H. The information rate ρ of the TUM scheme based on r-uniform hypergraph prohibited structure:

(1) $1/(C(n-1, r-1) - d + 1) \le \rho \le 1/ (C(n-1, r-1) - d)$ if $d \ge r$,

(2) $\rho = 1/ (C(n-1, r-1) - d)$ if $d < r$,

(3) $\rho \ge 1/ C(n-1, r-1)$.

Proof. Because based on r-uniform hypergraph prohibited structure, $\max_{B\in\Delta} |B| = l = r$. Let deg($p_i$) denote the degree of p_i.

(1) Each participant p_i must get $C(n-1, r-1) - \deg(p_i)$ shares at the step 3 in Sub-section 4.1. Because $d \ge r$, there may some participants in T which is definition in Sub-section 3.4. Hence, these participants who are in T must get one secret at the step 2 in Sub-section 4.1. every participant p_i may get $C(n-1, r-1) - \deg(p_i)$ shares (if he(she) is not in T) or $C(n-1, r-1) - \deg(p_i) + 1$ shares (if he(she) is in T). Hence, $1/(C(n-1, r-1) - d + 1) \le \rho \le 1/ (C(n-1, r-1) - d)$ if $d \ge r$.

(2) Because $d < r$, there is no participants in T. Hence, each participant p_i must get $C(n-1, r-1) - \deg(p_i)$ shares at the step 3 in Sub-section 4.1. Hence, $\rho = 1/ (C(n-1, r-1) - d)$ if $d < r$.

(3) It is easy to see by (1) and (2). □

Corollary 4.2.
The lower bound of the information rate of the TUM perfect secret sharing scheme is $\max\{1/(C(n-1, r-1) - d + 1), 1/(C(n-1, r-1))\}$, where d is the minimum degree of H and n is the number of vertices of H.

We compare our scheme with TUM scheme and list the result in Table 3, where n is the number of participants and d is the minimum degree of H.

Table 3: The Information Rate of Our Scheme and TUM Scheme

	Information rate
r-HP scheme	$\max\{r/(C(n-1, r-1) - d + r + 1), r/(C(n-1, r-1) + 1)\}$
TUM scheme	$\max\{1/(C(n-1, r-1) - d + 1), 1/(C(n-1, r-1))\}$

5. Conclusion

In this paper, we given a new construction of perfect secret sharing scheme based on r-uniform hypergraph for prohibited structures, where a vertex denotes a participant and an edge is a set of r participants who cannot recover the secret. The information rate of our scheme is either $r/(C(n-1, r-1) - d + r + 1)$ or $r/(C(n-1, r-1) - d + 1)$, and we can show that the value is always greater then or equal to $r/(C(n-1, r-1) + 1)$, where d is the minimum degree of H. The r-HP scheme include the scheme held by Sun [19] as a special case for $r = 2$. And we can find that r-HP scheme is better than TUM scheme for any nontrivial cases (that means for $r > 1$ and $d > 0$).

Our *r*-HP scheme is more efficient than TUM scheme, but TUM scheme is based on general access structure. We expect that our scheme can be extend to solve the general hypergraph. This problem is as same as the problem of secret sharing scheme for general access structure. And that is our future work.

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